

AREA III:

AB Structures and Environment Engineering and Bioprocess Engineering and Allied Subjects

TOPIC 1:

Agricultural Buildings and Structures

1.5 Apply the theories and principles of psychrometric in the design, construction, operation, and management of greenhouses and appurtenant structures and facilities for the production of high value crops.

Number of items: 5-6 questions

Coverage: Psychrometry

Psychrometry

This deals with the thermodynamic properties of mixed air and water vapor, often used in HVAC (heating, ventilation, and air conditioning) systems to control environmental conditions.

In the context of greenhouses and appurtenant structures, applying psychrometric principles involves managing temperature, humidity, and air quality to create optimal growing conditions for high-value crops.

Psychrometry

This involves calculating the dew point, humidity ratio, and enthalpy to maintain a conducive environment for plant growth. This also involves the monitoring of air's dry-bulb temperature, wet-bulb temperature, and humidity levels.

In simple words, psychrometry is the study of **MOIST AIR or mixture of dry air and water vapor.**

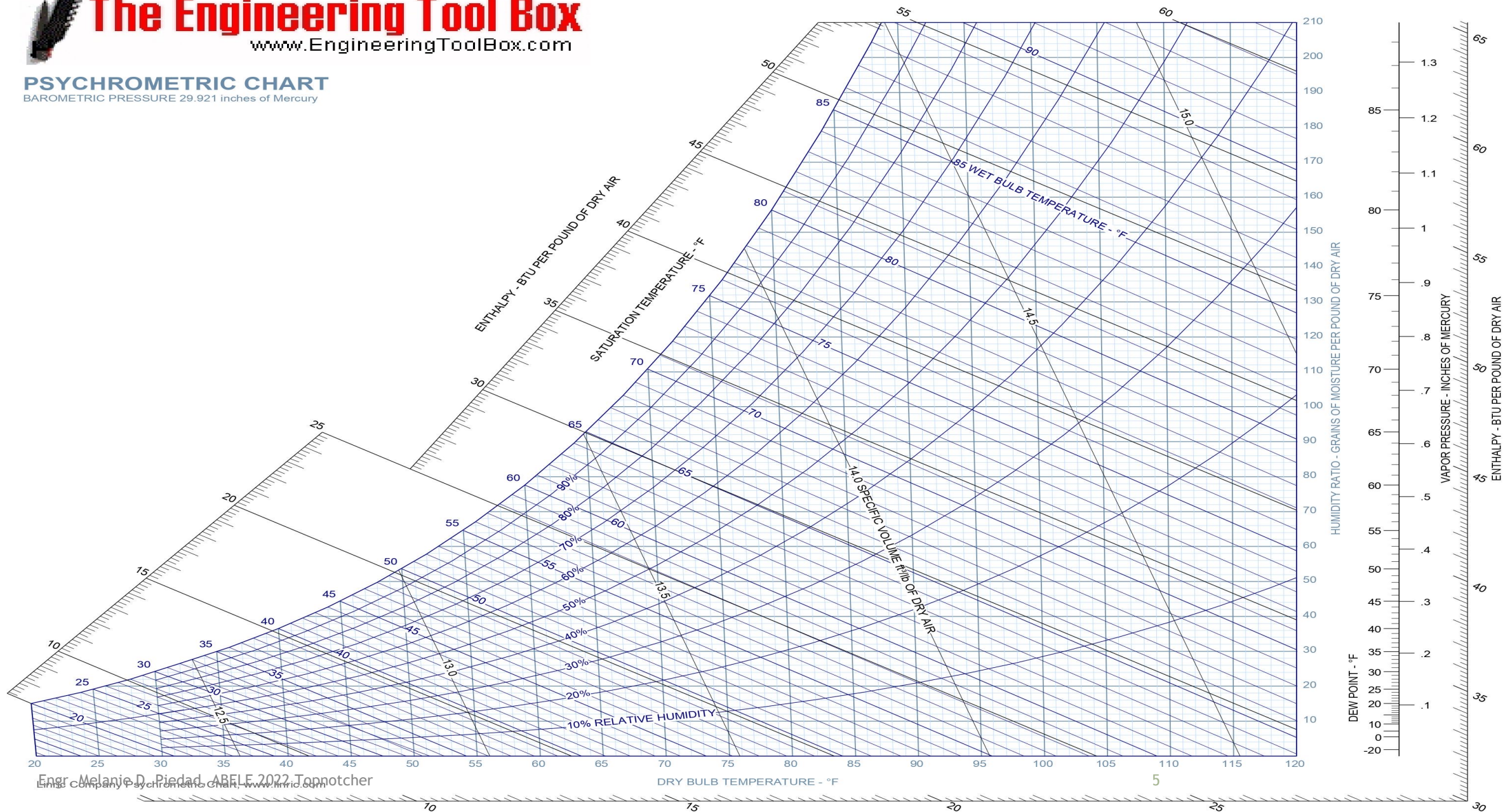


The Engineering Tool Box

www.EngineeringToolBox.com

PSYCHROMETRIC CHART

BAROMETRIC PRESSURE 29.921 inches of Mercury



Engr. Melanie D. Piedad, ABLE 2022 Topnotcher

Link Company Psychrometric Chart, www.lmc.com

Properties of Psychrometry

- Dew point temperature
- Relative humidity
- Humidity ratio
- Dry-bulb temperature
- Wet bulb temperature
- Humidity (Specific humidity and Absolute humidity)
- Psychrometric ratio

Properties of Air

- Dry-bulb temperature
- Wet-bulb temperature
- Dew-point temperature
- Relative humidity
- Humidity ratio

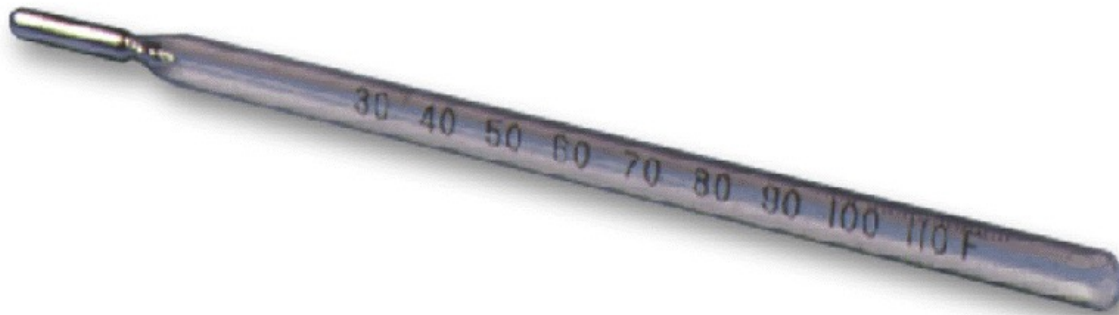
NOTE:

Condensation occurs at dew point

Fog occurs when air is saturated

Properties of Air

Dry-bulb temperature - it is the temperature of the air measured by a standard thermometer exposed to the air but shielded from moisture and radiation. It is called "dry bulb" because the thermometer bulb is dry and is the most commonly referred to air temperature.



Properties of Air

Wet-bulb temperature - it is the lowest temperature that can be achieved by evaporative cooling of a water-saturated surface. It is measured using a thermometer with a bulb wrapped in a wet cloth (wet bulb), over which air is passed. This temperature is an indicator of the moisture content in the air and is always lower than or equal to the dry bulb temperature.



Properties of Air

Dew Point Temperature - temperature at which the water vapor in the air is saturated.

Relative Humidity compares the moisture content to saturation

Humidity - water vapor in the air.

Absolute Humidity - also called as vapor density. This is the mass of water vapor per unit volume of air.

Properties of Air

Relative Humidity - ratio of the actual partial pressure exerted by the water vapor in any volume of air to partial pressure that would be exerted by the water vapor.

$$\text{Relative humidity} = \frac{\text{Actual partial pressure}}{\text{Partial pressure at saturation}} \times 100$$

Properties of Air

Humidity ratio - also called as specific humidity. This is the mass of water vapor per unit mass of dry air.

$$\text{Humidity Ratio} = (0.6222) \left(\frac{P_p}{P_t - P_p} \right)$$

Where

P_p = water vapor pressure of air (partial pressure)

P_t = atmospheric pressure (total pressure)

Other important Formula:

- **Sensible Heat Ratio (SHR)**

$$\text{SHR} = \frac{\text{Sensible Heat Gain}}{\text{Sensible Heat Gain} + \text{Latent Heat Gain}}$$

- **Supply Airflow (SA)**

$$\text{SA} = \frac{\text{Sensible heat gain}}{1.085 \times (\text{Room DB} - \text{Supply DB})}$$

- **Enthalpy (h)**

$$h = \text{Sensible heat} + \text{Latent heat}$$

Other important Formula:

- **Refrigeration Load (RL)**

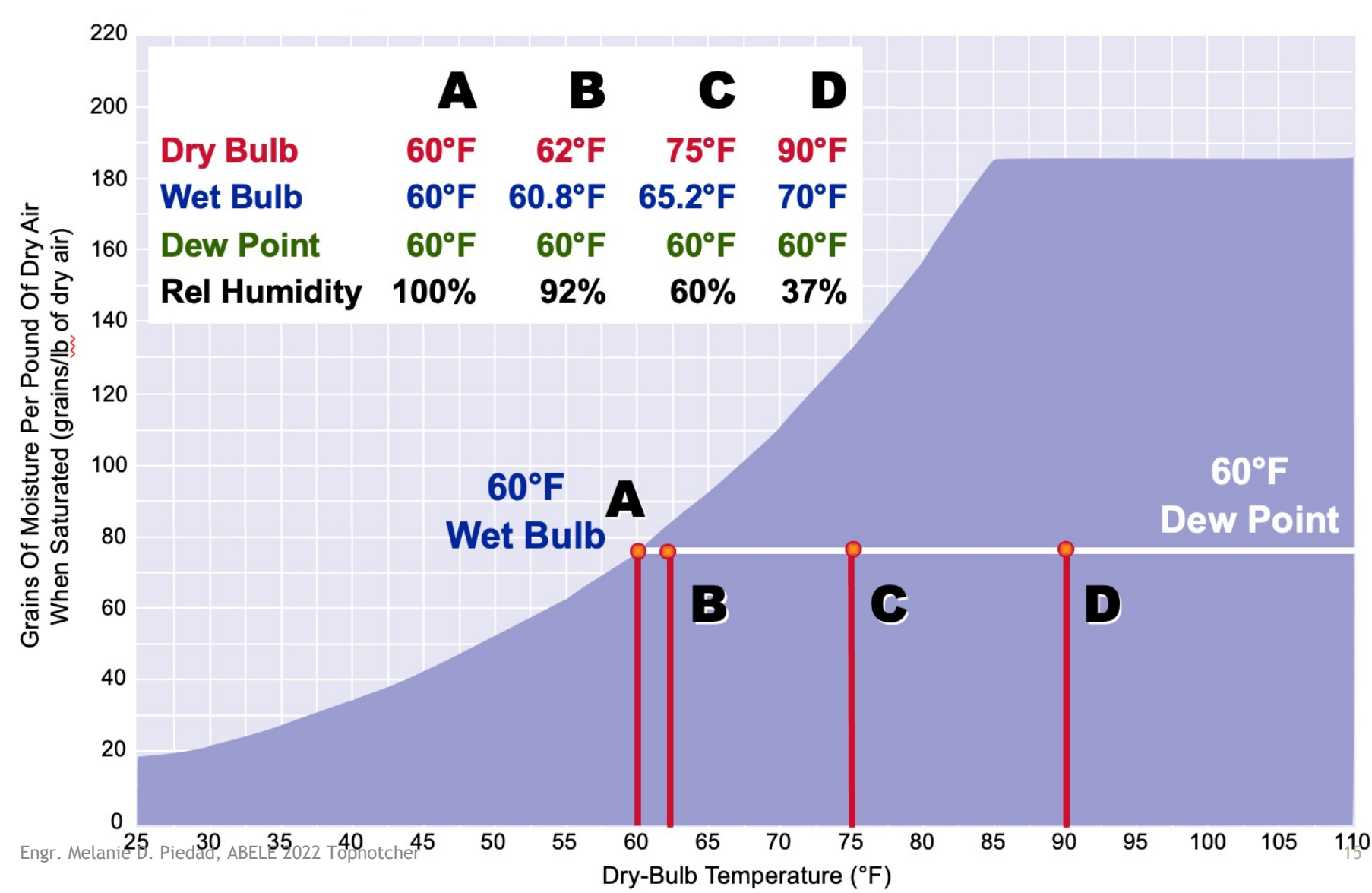
$$RL = 4.5 \times SA \times (h_1 - h_2)$$

where:

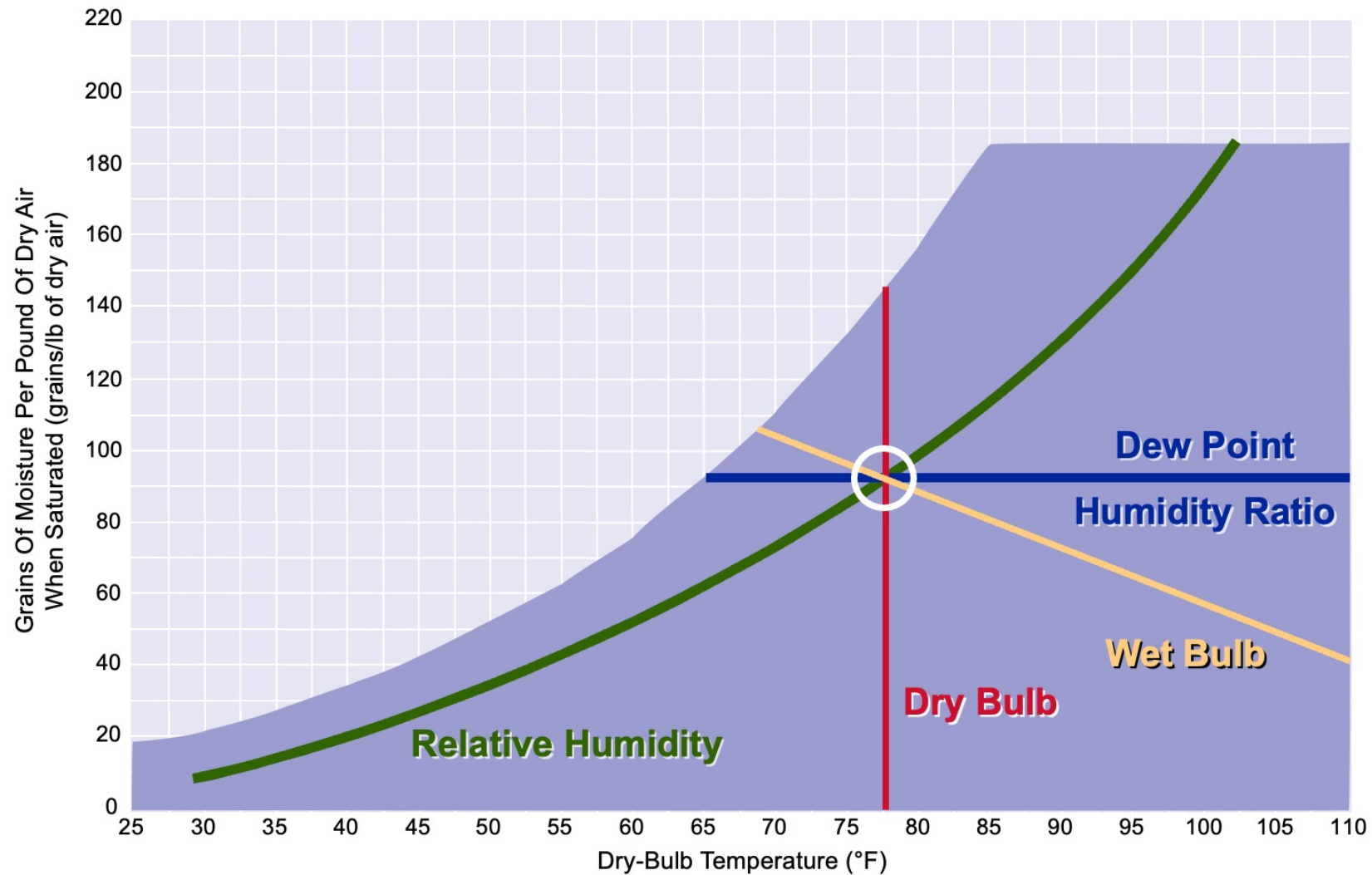
h_1 = enthalpy of air entering coil (Btu/lb)

h_2 = enthalpy of air leaving coil (Btu/lb)

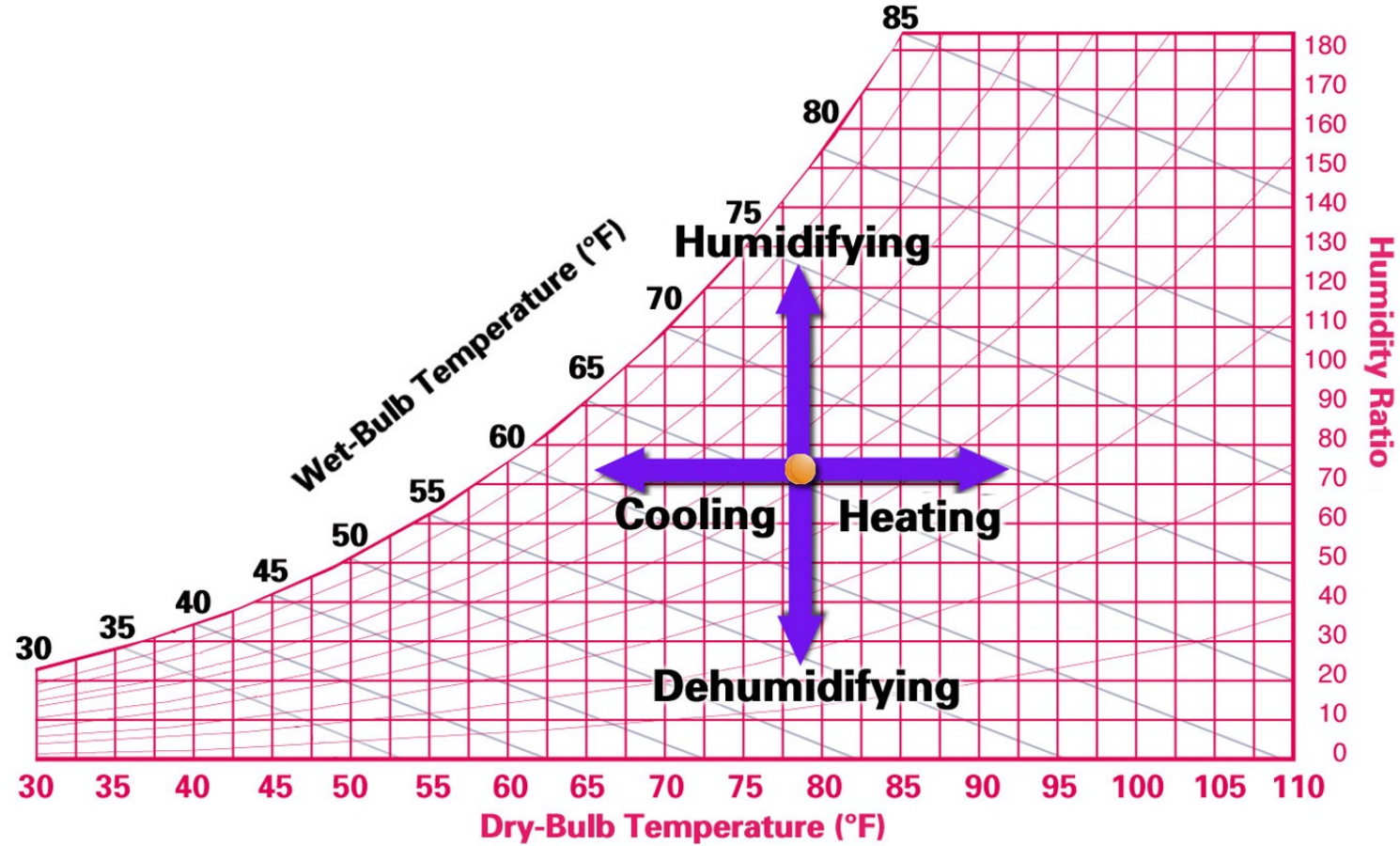
Relationship between Dry Bulb, Wet Bulb And Dew Point



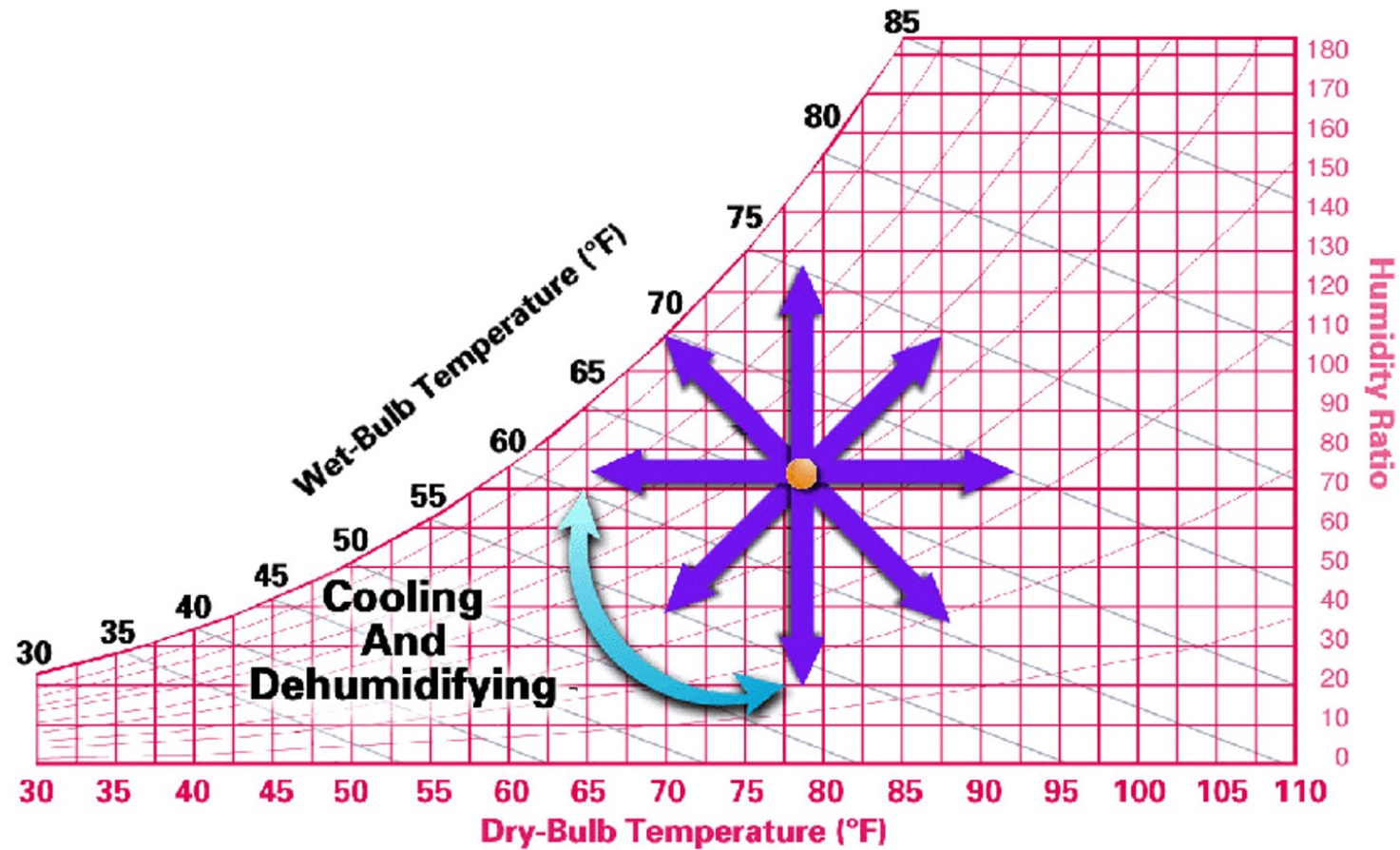
Intersection of Air Properties



Effects of changing sensible heat and moisture content

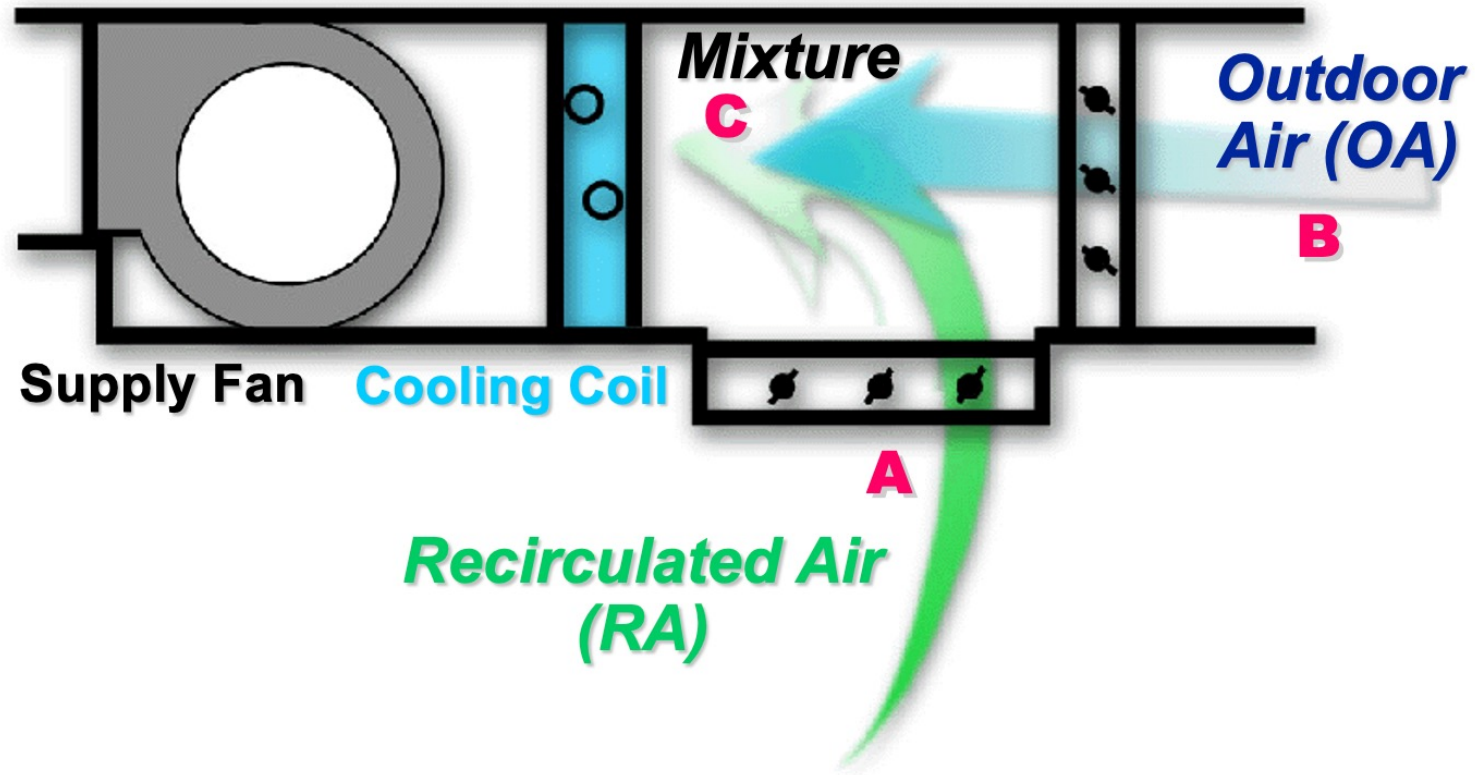


Effects of changing sensible heat and moisture content

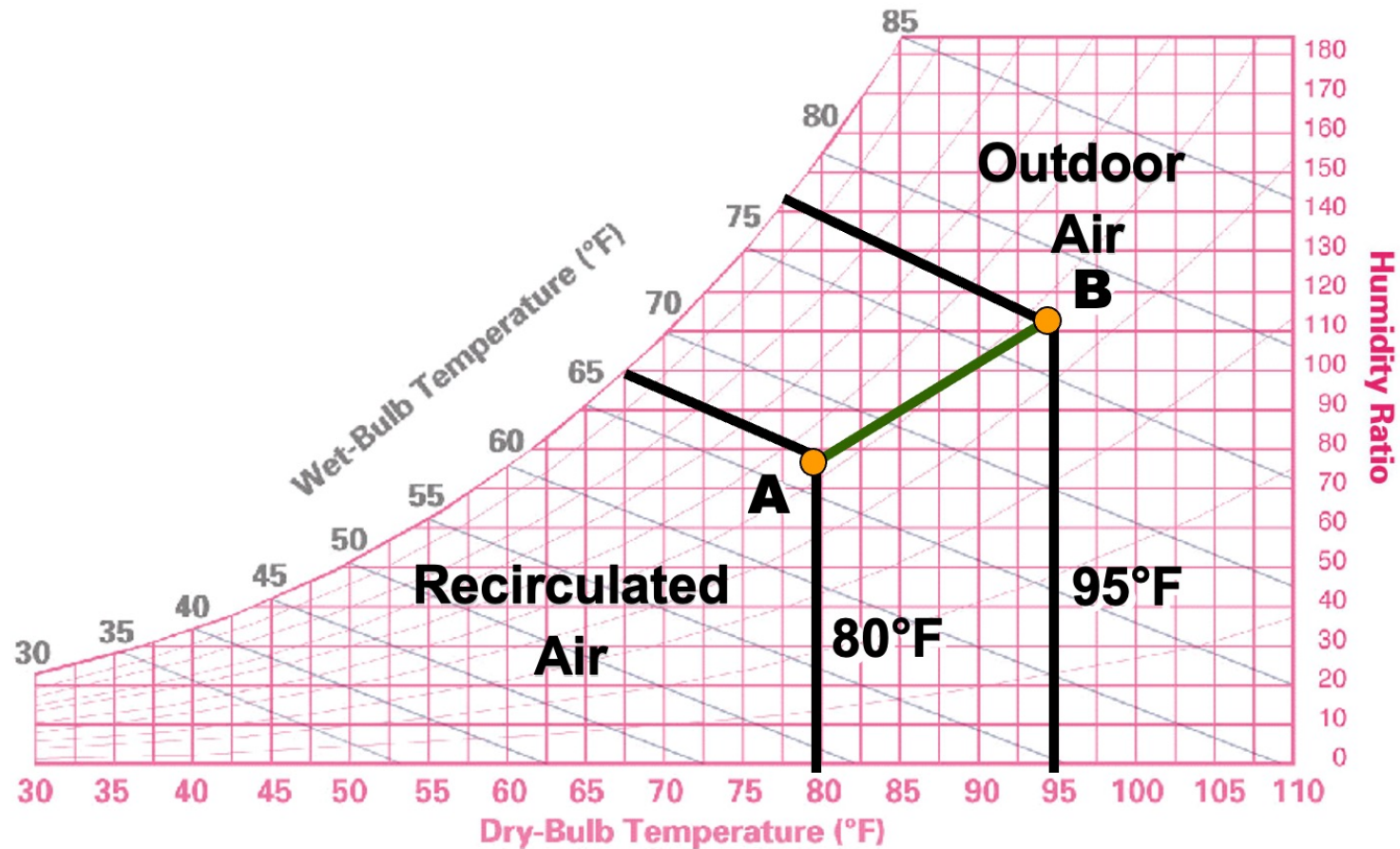


Air Mixture

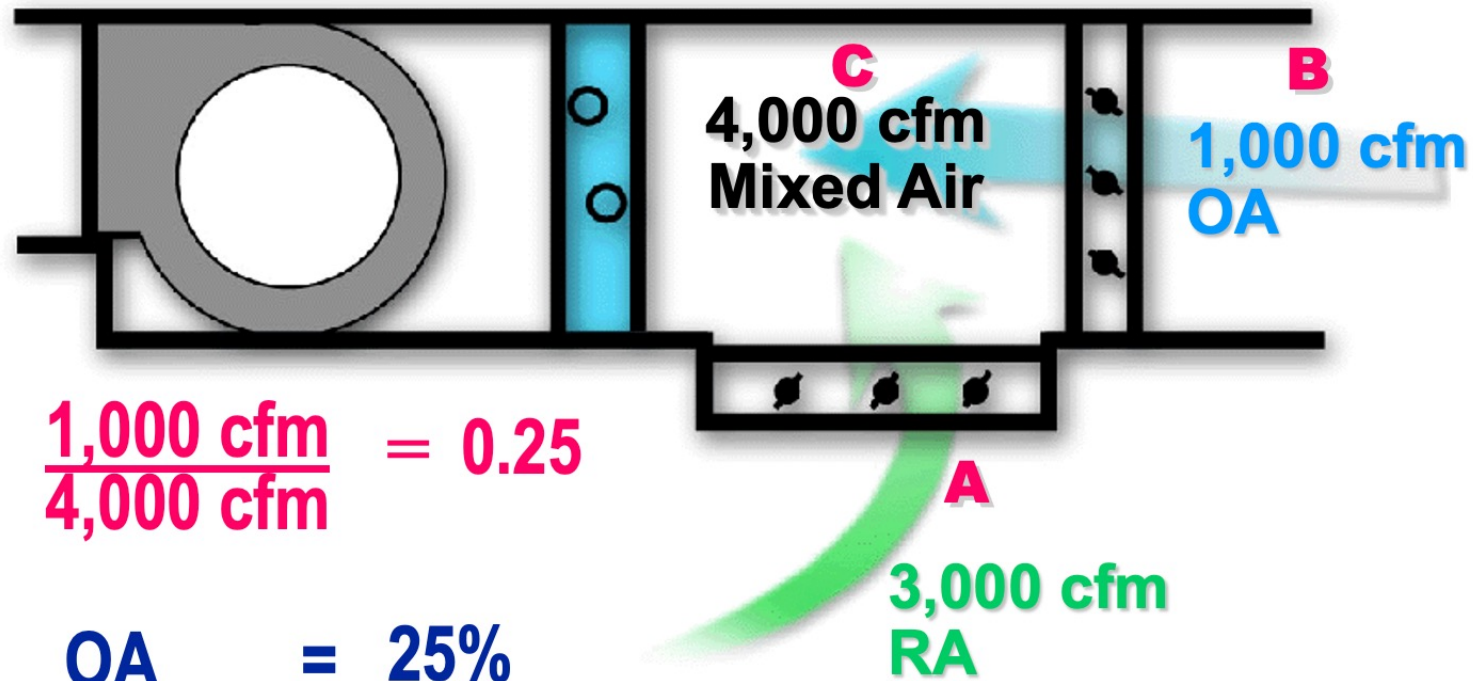
Air Conditioner



Sample questions



Sample questions



$$\frac{1,000 \text{ cfm}}{4,000 \text{ cfm}} = 0.25$$

$$\text{OA} = 25\%$$

$$\text{RA} = 75\%$$

$$\text{Mixture} = 100\%$$

Sample questions

Therefore, the mixed air temperature is:

$$95^{\circ}\text{F} \times 0.25 = 23.75^{\circ}\text{F}$$

$$80^{\circ}\text{F} \times 0.75 = 60.00^{\circ}\text{F}$$

$$\text{Mixture} = 83.75^{\circ}\text{F}$$

Sample questions

An outdoor and recirculated air temperature and flow is given to be 25°C and 98°F; 987 and 2312 cfm, respectively. What is the mixed air temperature (R) entering the room?

note - edit the choices

a. 549

b. 559

c. 79

d. 89

$$\frac{987}{2312 + 987} = 0.30$$

$$OA : 77 \times 0.30 = 23.1$$

$$RA : 98 \times 0.70 = 68.6$$

$$91.7^\circ\text{F} + 460 = 551.7\text{R}$$

Sample questions

$$MA_T = 33^\circ\text{C} = 91.4^\circ\text{F}$$

$$RA_T = 99^\circ\text{F}$$

$$OA_T = 29^\circ\text{C} = 84.2^\circ\text{F}$$

$$RA_F = 2500 \text{ cfm}$$

Ms. Irene is preparing the lecture room before class. A quick check shows the mixed air right before the cooling coil is 33°C . The recirculating air coming from the hall is 99°F at 2500 cfm, and the outdoor air is mild at 29°C . Assume steady mixing of dry air and no duct gains/losses. Compute for the OA flowrate being brought in.

a. 1240 cfm

b. 1350 cfm

c. 1532 cfm

d. 1420 cfm

$$\textcircled{1} \frac{OA_F}{OA_F + RA_F} = OAR$$

$$\textcircled{2} OAR + RAR = 1$$

$$OAR = 1 - RAR$$

$$\begin{array}{c|c} 84.2 \times (1 - RAR) & X \\ 99 \times (RAR) & Y \end{array}$$

$$MA_T = 91.4^\circ\text{F}$$

$$[(84.2)(1 - RAR) + (99)(RAR) = 91.4]$$

$$RAR = 0.49$$

$$OAR = 0.51$$

Sample questions

$$\frac{DA_F}{DA_F + 2500} = 0.51$$

$$DA_F = 2602 \text{ cfm}$$

At what Dry-bulb and wet-bulb temperature does the relative humidity become 100%, respectively?

- a. 60% and 50%
- b. 60% and 60%
- c. 50% and 50%
- d. 50% and 60%

Sample questions

In the psychrometric chart, what does the diagonal line represents?

- a. Wet-bulb temperature
- b. Dew point temperature
- c. Relative humidity
- d. Dry-bulb temperature

THANK YOU!!!

Proverbs 3:5-6

**“Trust in the Lord with all your heart
and lean not on your own
understanding; in all your ways
submit to Him, and He will make
your paths straight”**